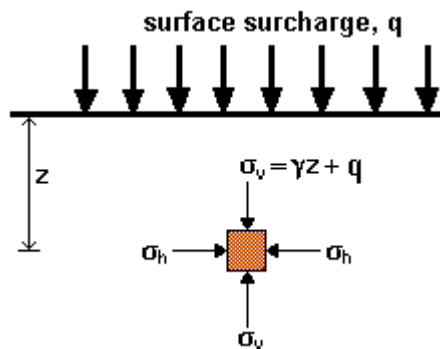


Section-A (25 Marks)

1. Define total stress and effective stress at a point below the ground level with an example. Briefly mention about the implication of total stress and effective stress at a point below the ground level with regards to design and construction of retaining wall. [2+3=5]

Answer:



Total stress (σ):

- Total stress at a point below the ground surface is the stress produced by the weight of all materials (soil, water, structures, etc.) above that point per unit area.
- It is the sum of surcharge and overburden pressure.

$$\sigma = q + \sum \gamma \times z$$

Effective stress (σ'):

- Effective stress is the portion of the total stress that is carried by the soil skeleton (soil particles to soil particles contact).
- According to the Terzaghi Effective Stress Principle,

$$\sigma' = \sigma - u$$

where: u = pore water pressure (kN/m^2)

σ' = effective stress (kN/m^2)

σ = total stress (kN/m^2)

u = pore water pressure (kN/m^2)

Example: Let us consider a saturated soil layer having:

Unit weight of saturated soil, $\gamma_{sat} = \frac{20 \text{ kN}}{\text{m}^3}$

Water table at ground surface

Depth of point below ground surface, $z = 5 \text{ m}$

Step 1: Calculate Total Stress

$$\sigma = \gamma_{sat} \times z$$

$$\sigma = 20 \times 5 = 100 \text{ kN/m}^2$$

Step 2: Calculate Pore Water Pressure

$$u = \gamma_w \times z$$

Taking unit weight of water $\gamma_w = 9.81 \text{ kN/m}^3$,

$$u = 9.81 \times 5 = 49.05 \text{ kN/m}^2$$

Step 3: Calculate Effective Stress

$$\begin{aligned}\sigma' &= \sigma - u \\ \sigma' &= 100 - 49.05 \\ \sigma' &= 50.95 \text{ kN/m}^2\end{aligned}$$

Concepts of total stress and effective stress are very important in geotechnical engineering, especially for the design and construction of retaining walls.

1. Implication of Total Stress

- Total stress represents the combined effect of the weight of soil and water above a point.
- The total lateral pressure acting on a retaining wall includes both soil pressure and water pressure.
- Neglecting total stress may lead to underestimation of the forces acting on the wall, causing instability.

2. Implication of Effective Stress

- Effective stress governs the shear strength, bearing capacity, and deformation of the retained soil.
- Earth pressure theories such as Rankine's and Coulomb's are generally based on effective stress parameters (c' and ϕ').
- The stability of retaining walls against sliding, overturning, and bearing failure depends largely on the effective stress in the soil.
- Change in groundwater level affect pore water pressure, which in turn changes effective stress and the lateral earth pressure on the wall.
- Failure to account for pore water pressure may result in excessive wall movement, cracking, or collapse.

Both total stress and effective stress must be considered in the design of retaining wall. Total stress determines the overall load acting on the wall, while effective stress controls the soil strength and stability of the wall-soil system. Therefore, safe and economical retaining wall design requires careful evaluation of both stresses and proper groundwater drainage provisions.

2. While designing a reinforced concrete section for bending, according to the area of reinforcement provided there are three types of sections. How these sections are classified and explain which one is most preferable and why. [5]

Answer:

According to the area of reinforcement provided while designing a reinforced concrete section for bending there are three types of sections. They are:

- i. Balanced section
- ii. Under-reinforced section
- iii. Over reinforced section

i. Balanced section:

- The section in which maximum stress in the concrete reaches at the same time as that of steel is known as balanced section.
- In this type of section concrete and steel fails simultaneously.
- Balanced section is an ideal section and does not exist in practice.
- The area of steel provided is exactly equal to the balanced reinforcement area.
- **Condition:**

$$A_s = A_{sb}$$

ii. Under-reinforced section:

- The section in which maximum stress in the steel reaches before that of concrete is known as under-reinforced section.
- In this type of section, steel fails before concrete.
- Depth of neutral axis is less than that of balanced section.
- Failure of under reinforced section is ductile or tensile in nature.
- The area of steel provided is less than the balanced reinforcement area.
- **Condition:**

$$A_s < A_{sb}$$

where:

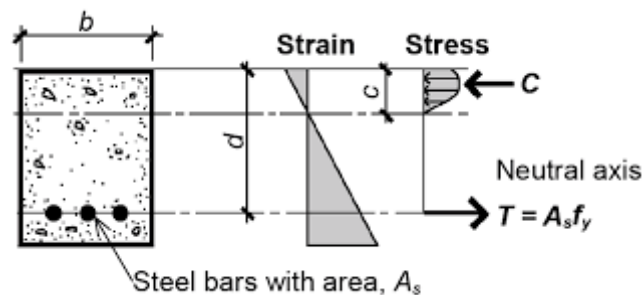
A_s = area of steel provided

A_{sb} = balanced area of steel

iii. Over-reinforced section:

- The section in which maximum stress in the concrete reaches before that of steel is known as over reinforced section.
- In this type of section, concrete fails before steel.
- Depth of neutral axis is higher than that of the balance section.
- The area of steel provided is greater than the balanced reinforcement area.
- Failure is sudden and brittle with little warning.
- **Condition:**

$$A_s < A_{sb}$$



Among all these three sections, the most preferable section is under reinforced section because:

- a. Failure in under-reinforced section is ductile or tensile in nature where large deflection and cracks occur before failure giving sufficient warning about the occurred failure.
- b. Under-reinforced section gives sufficient time for the occupant to vacate the structure and take precautionary measures before complete failure occurs.
- c. Moment of resistance (MOR) increases in under reinforced section.
- d. Long term and short-term deflection reduce.
- e. Under-reinforced section is generally recommended by design codes.

3. Define the normal stress, shear stress and principal stresses. Using a neat Mohr circle, show how normal and shear stress are obtained from the principal stresses for a plane stress state. [4+6=10]

Answer:

Normal stress(σ):

It is the stress acting normal to the surface or plane under consideration.

It is produced by axial tensile or compressive forces.

$$\sigma = \frac{P}{A}$$

where:

σ = normal stress (N/m² or Pa)

P = normal force acting on the area

A = cross-sectional area

(b) Shear Stress (τ):

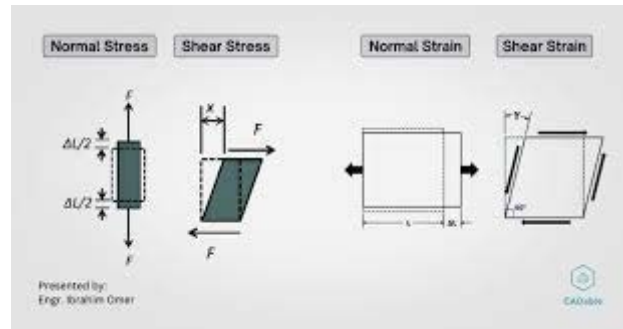
Shear stress is the stress acting **parallel (tangential)** to a surface or plane. It tends to cause one layer of material to slide over another.

$$\tau = \frac{V}{A}$$

where:

τ = shear stress (N/m² or Pa)

V = shear force
 A = area resisting shear



(c) Principal Stresses:

Principal stresses are the maximum and minimum normal stresses acting at a point on particular planes where the shear stress is zero.

They are given by:

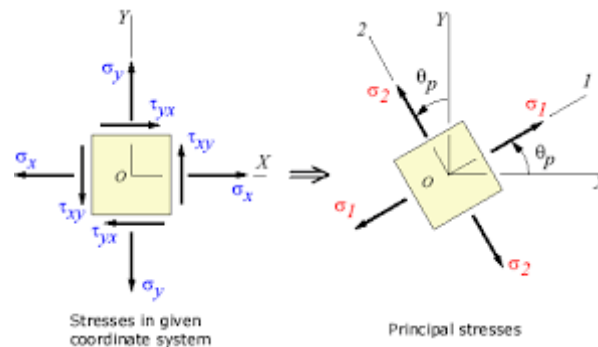
$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

σ_1 = Major principal stress (maximum)

σ_2 = Intermediate principal stress

σ_3 = Minor principal stress (minimum)

For a two-dimensional stress system:



Mohr's circle:

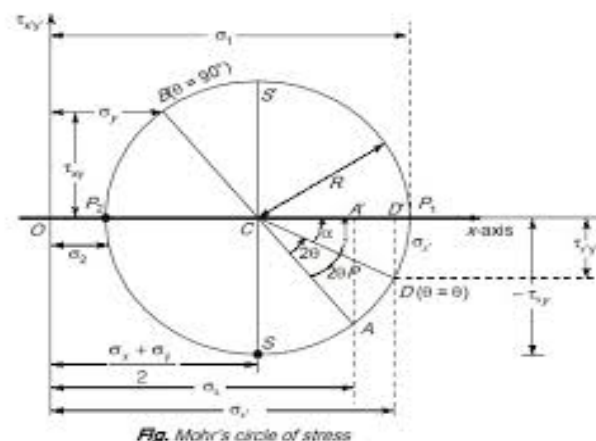
Consider a plane stress condition where the principal stresses are:

σ_1 = Major principal stress

σ_2 = Minor principal stress

On the principal planes, the shear stress is zero. Let a plane be inclined at an angle θ to the principal plane.

The stresses acting on this inclined plane can be obtained using Mohr's Circle.



Here, $\sigma_{avg} = \frac{\sigma_1 + \sigma_2}{2}$ and $R = \frac{\sigma_1 - \sigma_2}{2}$

Normal Stress on an Inclined Plane:

$$\sigma_n = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta$$

Shear Stress on an Inclined Plane:

$$\tau_n = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta$$

Maximum Shear Stress: Occurs when $2\theta = 90^\circ$

Hence, $\tau_{max} = \frac{\sigma_1 - \sigma_2}{2}$

2. **Principal Stresses:** Occur where $\tau = 0$ and are equal to σ_1 and σ_2

Using Mohr's circle, the stresses on any plane inclined at an angle θ to the principal plane are:

$$\sigma_n = \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta$$

$$\tau_n = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta$$

These equations show how the **normal stress** and **shear stress** on any plane are obtained directly from the **principal stresses** σ_1 and σ_2 using Mohr's circle.

4. **Discuss the reasons behind failure of roads bridges. What are the challenges in road bridge maintenance? Explain in brief the specific design considerations and construction methodologies and process to avoid bridge failures. [3+2+5=10]**

Answer:

Road bridges are that structure designed to carry vehicular and pedestrian traffic across the physical obstacles to be crossed like valleys, rivers etc. In the country like Nepal with large number of drainages and rugged topography, road bridges are common structures required for crossing rivers.

Road bridges failures are those events that causes severe damage or collapse of road bridges primarily driven by human errors or natural forces. The reasons behind failure of road bridges are discussed in points below:



- i. **Hydraulic and natural forces:**
Bridge failures commonly occur due to flooding, scouring or other natural calamities like earthquake, landslides, hurricanes.
In regions like Nepal, water-induced damage accounts for nearly 40% of failures.
- ii. **Structural overloading:**
Bridge failure also can occur due to structural overloading where vehicular load exceeds the load limits.
- iii. **Construction defects:**
Poor workmanship, inferior construction materials, no quality control can also lead to bridge failures.
Failure of Road bridges due to faulty construction accounts for 28%.
- iv. **Design flaws:**

Incorrect design, miscalculated load, inadequate foundation depth results in road failures.

Some of the popular examples of road bridge failures:

- i. **Tacoma Narrows suspension bridge** collapsed into the Tacoma Narrows strait just four months after opening due to strong winds causing the slender structure of the bridge to twist violently and break apart.
- ii. **Jabdighat concrete bridge Bardiya:** constructed by Pappu Construction Company, spanning over Babai River suffered severe foundation settlement and structural collapse during intense floods due to scour and insufficient geotechnical investigation.
- iii. **An under-construction motorable bridge collapsed**, connecting Thimura of Chitwan district and Devghat of Tanahun district.

Road bridge maintenance faces multiple challenges. Some of them are mentioned in points:

- i. **Lack of skilled personnel:**
 - Shortage of skilled manpower required for bridge maintenance
 - Insufficient training and expertise in modern maintenance.
- ii. **Insufficient funding for maintenance:**
 - Limited budget is challenging to allocate resources effectively for all bridge maintenance.
 - Lack of continuous funding for bridge maintenance program
- iii. **Traffic disruption:**
 - Performing repairs on busy arterial bridges requires lane closures causing significant congestion, economic disruption, and public frustration.
- iv. **Planning and operating issues:**
 - Poor project selection and prioritization
 - Lack of records and coordination.

Specific design considerations to avoid bridge failures are described in brief:

- i. **Determination of load:**

Road bridges are subjected to heavy vehicular traffic in addition to their own self-weight. Therefore, they must be designed to safely withstand dead loads, live loads, impact loads, and other applicable forces throughout their service life.
- ii. **Hydraulic and geotechnical safety:**

Road bridges must be designed to withstand extreme hydrological, geological and climatic conditions to avoid failure of bridges due to extreme flood, landslides and scour.
- iii. **Ductility:**

Redundancy and ductility of bridge should be improved by proper detailing so that when one component of structure fails, it can be supported by another component.

Construction methodologies and process to avoid bridge failures:

- i. **Staged construction:**

Complex design and construction of the structures like road bridge require methodical erection to ensure that the structure is stable at every stage, preventing collapses during temporary support phase.
- ii. **Strict quality control:**

Strict material testing such as concrete slump, steel tension tests etc. must be carried out. Ensuring proper curing times, independent third-party design reviews help to mitigate poor workmanship.

Prestressing is one of the most important methods for preventing failure in concrete bridges. Plain concrete is weak in tension which easily forms cracks. These cracks lead to corrosion of reinforcement ultimately leading to the failure of bridge section. So, to overcome this type of failure, prestressing is introduced in the concrete where concrete is subjected to compression *before* service loads are applied, so that when bending loads are subjected, they have to overcome that compression before any tension or cracking develops.

There are two methods of prestressing:

- i. **Pre-tensioning:** usually done off-site in a precast plant maintaining very high-quality control, since strands are tensioned and concrete is poured in controlled factory conditions. It's used for standard precast girders, deck panels, and beams.
- ii. **Post-tensioning:** usually done on-site, especially for long-span bridges, box girders, and segmental construction.

Section-B (25 Marks)

5. In the context of open channel flow, define specific energy. Provide a free hand sketch that shows combined graphical variation of different components of specific energy and also label and explain different parameters and features. [5]

Answer:

Specific energy is defined as the energy at a cross-section of an open channel flow with respect to channel bed. It is the total energy per unit weight of water measured with respect to the channel bed as the datum at a particular section of an open channel.

It is the sum of:

- i. Depth of flow (potential energy), and
- ii. Velocity head (kinetic energy).

Mathematically,

$$E = y + \frac{V^2}{2g}$$

where:

E = specific energy (m)

y = depth of flow (m)

V = mean velocity of flow (m/s)

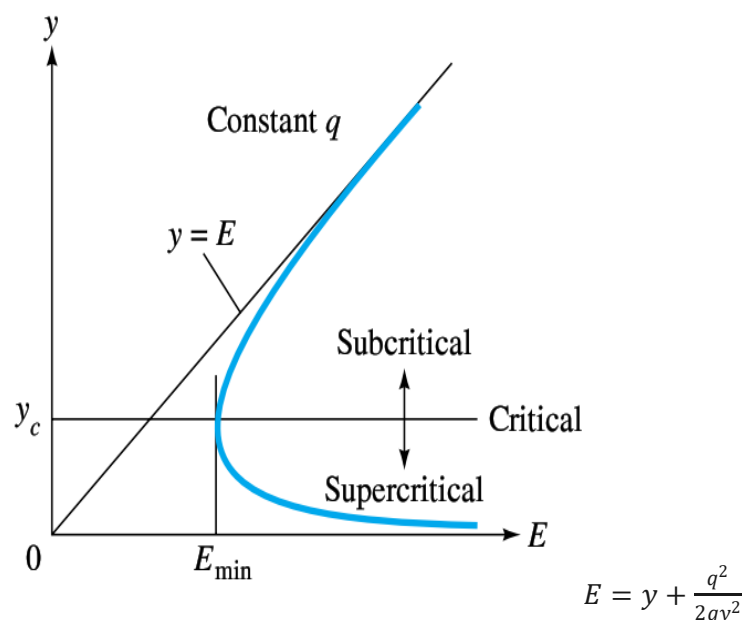
g = acceleration due to gravity (9.81 m/s²)

For a rectangular channel carrying a constant discharge per unit width (q):

$$E = y + \frac{V^2}{2g}$$

And,

$$V = \frac{q}{y}$$



Features:

- i. Different value of Q gives the different curve for specific energy.
- ii. For a particular discharge Q , there are two possible depths with same specific energy E_1 . These are alternate depths.
- iii. Two alternate depths represent completely different flow regimes- slow and deep on the upper portion of the curve and fast and shallow on the lower portion of the curve.
- iv. The upper regime of the flow with $y > y_c$ is in subcritical regime and lower regime of the flow with $y < y_c$ is in a supercritical regime.

Labeled Parameters:

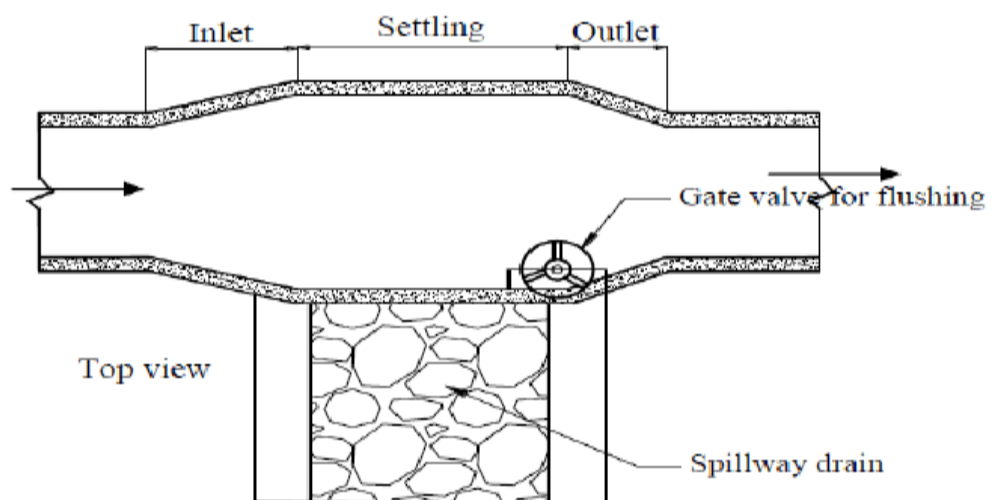
- a. **Depth of flow (y):** Plotted on the vertical y-axis.
- b. **Specific Energy (E):** Plotted on the horizontal x-axis.
- c. **Minimum Specific Energy (E_c):** The absolute lowest point on the C-shaped curve where the specific energy is minimized.
- d. **Critical Depth (y_c):** The exact depth of flow that corresponds to E_c .
At this point, $Fr=1$ i.e. flow transitions between regimes.
- e. **Alternate Depths:** For any given specific energy greater than E_c . There are two distinct depths that can carry the same discharge: one is deep with a low velocity, and the other is shallow with a high velocity.
- f. **Subcritical Flow Limb:** The upper portion of the specific energy curve (above critical depth (y_c)) where depths are large and velocities are slow i.e. $Fr < 1$.
- g. **Supercritical Flow Limb:** The lower portion of the curve (below critical depth (y_c)) where depths are shallow and velocities are rapid ($Fr > 1$).

6. Discuss the design considerations for a de-sanding basin in a hydropower project with sketch. Explain the importance of considering suspended sediment characteristics in the design. How are sediments flushed from a de-sanding basin. [4+3+3=10]

Answer:

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between intake and penstock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.

The design considerations for sediment flushing in the de-sanding basin are listed below:



i. Flushing channel:

- Flushing channels must be provided at the bottom of settling basin for the efficient collection and removal of sediments.

ii. Flushing velocity:

- The flushing velocity must be high enough for the transportation of settled sediment out of settling basin.
- Usually, flushing velocity of 2-3 m/s must be maintained during flushing.

iii. Bottom slopes:

- Adequate longitudinal and transverse slope are provided to facilitate efficient movement of sediments towards flushing outlet.

iv. Flushing gates:

- Flushing gates should be located at the suitable positions and designed to pass sediment-laden water quickly and efficiently.

v. Water requirement:

- Sufficient water discharge is required to generate required flushing velocity.

The importance of considering suspended sediment characteristics in the design are explained in points below:

i. Reservoir sedimentation:

When the flow velocity in reservoirs and dams reduces, the suspended sediments begin to settle at the bottom. This sedimentation of suspended sediments reduces storage capacity of reservoir over time. Hence, Proper assessment of sediment characteristics helps to estimate the rate of sediment deposition and determine the useful life of the reservoir.

ii. Prevention of wear and tear of hydraulic structures:

Sediment-laden water causes abrasion of turbines, pumps, gates, spillways, and canal linings. Understanding sediment characteristics allows engineers to select suitable materials and protective measures to minimize wear and maintenance costs.

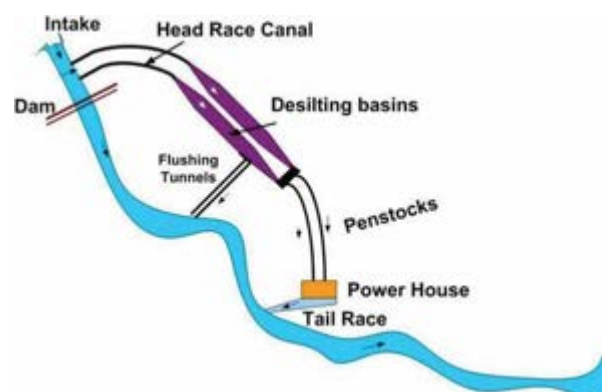
iii. Water quality management:

As the concentration of the suspended sediment is high, turbidity of water increases which reduce water quality and affect aquatic ecosystems. Hence, water treatment facilities must be adequately designed to ensure effective treatment and supply of clean water.

iv. Canal design and river training works:

Sediment deposition reduces the carrying capacity of canals and alter river flow patterns. Hence, understanding sediment characteristics helps in designing canals and river training works to prevent silting and flow obstruction.

Process of flushing the sediments from de-sanding basin:



i. Sediment Settlement:

The velocity of flow gets reduced when water enters the de-sanding basin due to transition at the inlet. This allows the coarse suspended sediments to settle at the bottom by the action of gravity while clear water flows out of the basin.

ii. Opening of flushing gates:

Flush gates are provided at downstream or along the bottom of the basin. In intermittent type of de-sanding basin, flush operation is carried out when the basin is filled with sediments. In continuous type, flush operation is conducted without obstruction.

iii. High velocity flow in the channel:

Opening of flush gates increases the velocity of the flow through the channel which gives enough tractive force to carry deposited sediments.

iv. Transportation of sediments and disposal:

The flushed sediments are transported through flushing channels and discharged back to river.

After flushing, the gates are closed and normal operation resumes.

7. What do you mean by river training works? Introduce the different methods of river training. [2+8=10]

Answer:

River training works are the structural engineering measures constructed along rivers to control the flow, stabilize the channel, and prevent bank erosion. They are designed to guide water within a specific path, safely pass floodwaters, and protect nearby infrastructure from shifting currents.

In Nepal, the rivers of terai and inner terai plains are carried out with extensive river training works because of the high sedimentation and unpredictable flooding during the monsoon season. Koshi river which brings the second largest sediment deposition among all the rivers in the world is highly prone to frequent flooding and lateral shifting. Several projects are carried out to construct massive embankments and guide bunds to protect settlements and agricultural lands near Koshi river and its tributaries.

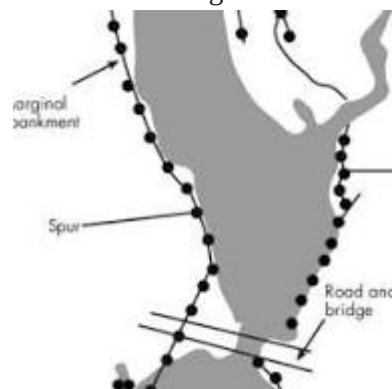
Depending upon the purpose required river training works can be classified into three types.

- i. high water training works: training for discharge
- ii. low water training works: training for depth
- iii. medium water training works: training for sediment

There are different methods of river training works. Some of them are described below:

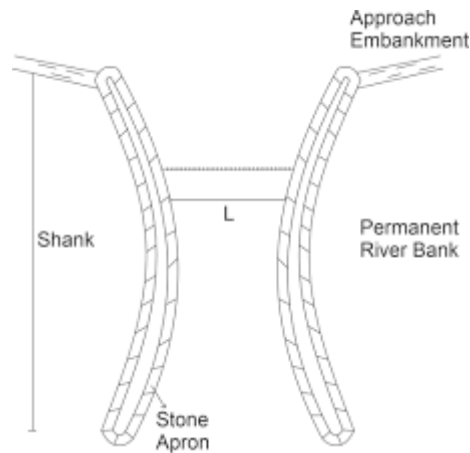
i. Marginal embankments/ levees:

- Marginal embankments are the earthen embankments constructed parallel to the riverbanks to confine flood waters within the cross section available between them.
- They are usually placed at some distance away from the riverbank so that area of the flow is increased.
- Generally required for meandering rivers.



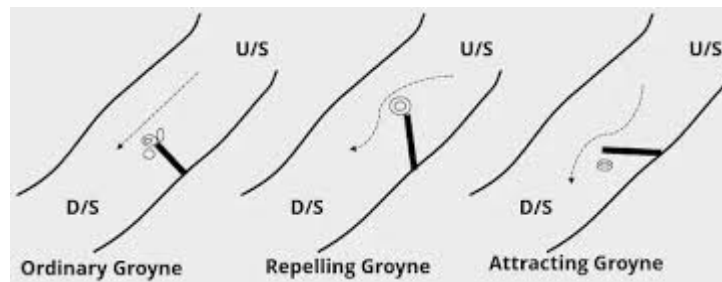
ii. Guide banks:

- Guide bank are the earthen embankments provided on one or both side of river at the bridge site to ensure that the flow path does not change the through the waterway at bridge site.
- The layout of guide banks should be such as to guide the flood smoothly through a work constructed across the river.



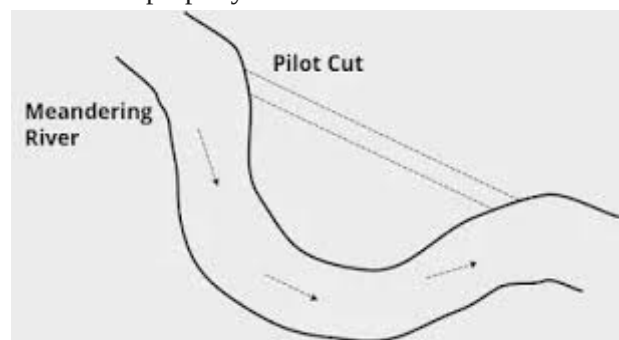
iii. Groynes or spurs:

- Groynes are the structure constructed transverse to the river flow and extend from the bank into the river.
- They help in training the river along a desired course by attracting, deflecting or repelling the flow in the channel.



iv. Artificial cutoff:

- An artificial cutoff is developed in a meandering river to divert the flow along a straight course.
- They are constructed for training of river when its mender goes on increasing and endanger some valuable land or property on the banks.

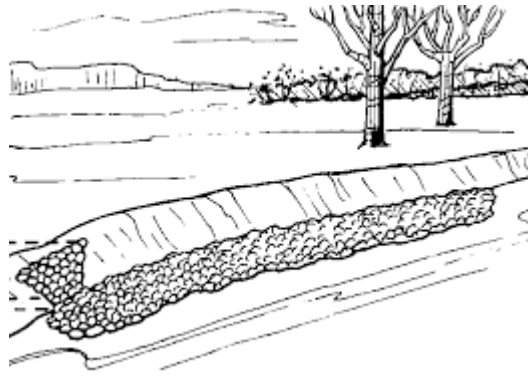


v. Bank protection and Pitched banks:

Bank protection helps in training of the river for the protection of hydraulic structures and adjacent land or property.

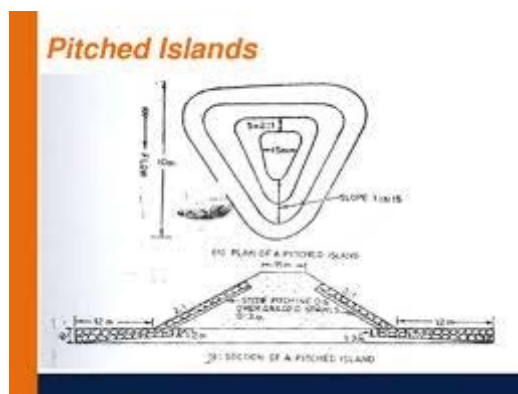
Bank protection work can be classified as direct or indirect.

Direct protection works include work done on the bank itself such as vegetation cover, pitching
Indirect protection includes work constructed not directly on the banks but in front of them.



vi. Pitched islands:

- Pitch Island is an artificially created island in the river bed, either made up of masonry or stone pitched on all sides.
- The main purpose of pitched island is to train the river to have an axial flow and improve the channel for navigation purpose.



vii. Bandalling:

- Bandalling is a method of river training works which is used to confine the low flow water of a shallow river in a single channel for maintaining the required depth for navigation.
- Bandalling started after the flood season is over and when the water level begins to fall.

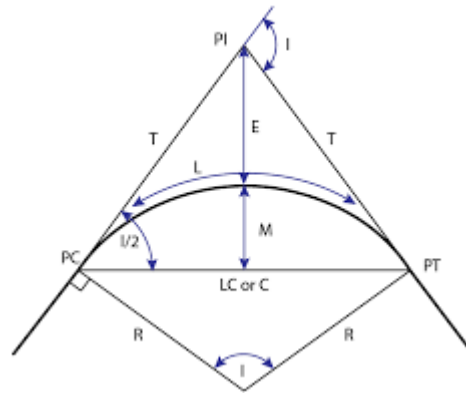


Section-C (25 Marks)

8. Discuss the design considerations associated with horizontal and vertical curves in highways. [5]

Answer:

A horizontal curve is a circular or transitional curve in a roadway alignment used to smoothly transition vehicles between two straight tangent sections. It enables gradual changes in direction, ensuring driver comfort, adequate sight distance, and safety while overcoming centrifugal forces.



Design considerations associated with horizontal curves in highways are discussed below:

i. Design Speed:

Horizontal curves are provided with larger radius of curvature for higher design speed. Sharp curves are avoided in the road designed for high speed.

ii. Radius of curvature:

The minimum radius is determined by:

$$R = \frac{V^2}{127(e + f)}$$

Where:

R = Radius of curve (m)
 V = Design speed (km/hr)
 e = Superelevation
 f = Coefficient of lateral friction

iii. Superelevation:

Super elevation is the transverse slope provided at the horizontal curve to counteract the centrifugal force by raising the outer edge of the pavement with respect to inner edge.

According to NRS 2070,

Maximum superelevation to be provided is limited to:

In plain and rolling terrain=7%

In snow bound areas=7%

In hilly areas not bound by snows=10%

Minimum value of super elevation should be equal to the rate of camber of pavement.

iv. Extra-widening:

Extra widening refers to the additional width of carriageway that is required on the curved section of a road over that required on a straight alignment.

The total extra widening is given by:

$$W_e = W_m + W_p$$

Where:

W_e = Total extra widening (m)

W_m = Mechanical widening (m)

W_p = Psychological widening (m)

Formula:

$$W_e = \frac{nl^2}{2R} + \frac{V}{9.5\sqrt{R}}$$

Where:

n = Number of traffic lanes

l = Wheel base of vehicle (m) (generally 6.1 m for design)

R = Radius of curve (m)

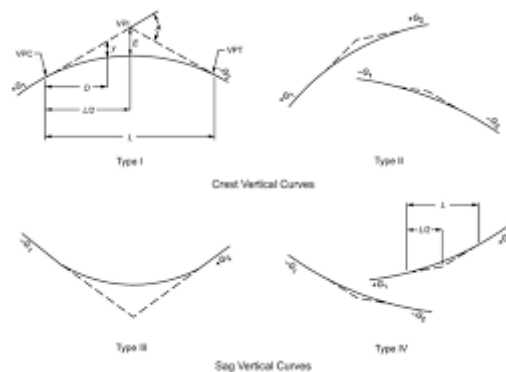
V = Design speed (km/hr)

v. **Sight distance:**

Sight distance is the actual distance along the roads which a driver from a specified height above the carriageway can see the objects either moving or stationary on the road surface.

Sight distances must be properly designed for efficient, economical and safe movement of vehicles on the road.

Vertical curves are parabolic shaped curves used in highway design to provide a smooth transition between two different roadway gradients (slopes). There are two types of vertical curves: Summit (Crest) Curve and Valley (Sag) Curve.



Design considerations associated with vertical curves in highways are discussed below:

i. **Rate of Change of Gradient:**

Change in gradient of the roads should be gradual to avoid discomfort and vehicle instability.

ii. **Sight Distance Requirement:**

Adequate stopping sight distance should be available throughout the curve for improved visibility and safety.

iii. **Length of Curve:**

Length of vertical curve depends on the difference between grades, design speed, required sight distances and so on.

iv. **Headlight Sight Distance (Valley Curve):**

Vehicle headlights should illuminate a sufficient distance ahead at night to reduce road accidents due to improper visibility and glare effect.

9. **What are the factors affecting stopping sight distance? Derive an expression to calculate stopping sight distance. [5+5=10]**

Answer:

Stopping sight distance (SSD) is the minimum sight distance available on the highway at any spot having sufficient length to enable the driver to stop a vehicle traveling at design speed safely without collision with any other obstruction.

In terms of speed in km/hr:

$$SSD = 0.278Vt + \frac{V^2}{254f}$$

Where:

V = Speed of vehicle (km/hr)

v = Speed of vehicle (m/s)

t = Reaction time (usually 2.5 s)
 f = Coefficient of longitudinal friction
 g = Acceleration due to gravity (9.81 m/s²)

There are several factors affecting the stopping sight distance some of which are listed below:

- i. **Speed of the vehicles:**
Higher the speed more will be the distance required to stop the vehicle. Hence, as the speed increases sight distance also increases.
- ii. **Efficiency of brakes:**
The efficiency of the brakes depends upon the vehicle characteristics, age of vehicles etc. If the brake efficient is 100%, the vehicle will stop at the moment the brake is applied. Hence, stopping sight distance will be more if the efficiency of brake is less.
- iii. **Frictional resistance between the tyre and the road:**
The frictional resistance between the tyre and the road brings the vehicle to stop. When the frictional resistance is more, the vehicle stops immediately which reduces the stopping sight distance.
- iv. **Gradient of the road:**
While moving in upward gradient, the vehicle can stop immediately so sight distance required is less but while descending a gradient, gravity also comes in action and more time is required to stop the vehicle.
- v. **Total reaction time of driver:**
Reaction time the driver is the total time taken from the instant object is visible to the driver to instant when the brake is applied. The reaction time of 2.5 seconds is considered in the design of stopping sight distance.

Derivation of expression to calculate stopping sight distance:

SSD consists of:

$$\text{SSD} = \text{Lag Distance} + \text{Braking Distance}$$

1. Lag Distance (Thinking Distance)

During the driver's reaction time t , the vehicle continues to move with speed v .

$$\text{Lag Distance} = vt$$

Where:

v = speed of vehicle (m/s)

t = reaction time (s)

2. Braking Distance

Using the equation of motion:

$$v^2 = u^2 + 2as$$

For a vehicle coming to rest:

Final velocity, $v = 0$

Initial velocity, $u = v$

Deceleration, $a = -fg$

Distance traveled, s = Braking Distance

Substituting,

$$0 = v^2 - 2fgs$$

$$s = \frac{v^2}{2fg}$$

Thus,

$$\text{Braking Distance} = \frac{v^2}{2fg}$$

Where:

f = coefficient of longitudinal friction

g = acceleration due to gravity (9.81 m/s²)

3. Total Stopping Sight Distance

$$SSD = vt + \frac{v^2}{2fg}$$

$$SSD = vt + \frac{v^2}{2fg}$$

When Speed is in km/hr

$$v = 0.278V$$

where V is in km/hr,

$$SSD = 0.278Vt + \frac{V^2}{254f}$$

$$SSD = 0.278Vt + \frac{V^2}{254f}$$

Where:

SSD = Stopping Sight Distance (m)

V = Speed of vehicle (km/hr)

t = Reaction time (usually 2.5 s)

f = Coefficient of longitudinal friction.

10. What are the factors affecting the length of an airport's runway? Explain the need for holding bays and bypasses. [5+5=10]

Answer:

The minimum length required for the takeoff of the aircraft at the maximum takeoff mass and standard atmospheric condition is known as airport runway length. There are several factors affecting the length of an airport's runway. They are as follows:

i. Aircraft characteristics:

- The length of runway is affected by aircraft performances, types of aircraft, takeoff and landing weight and so on.
- Large size of aircraft requires longer runways.
- Heavier aircraft with higher takeoff and landing weights requires longer runways to reach takeoff speed and to stop.
- The length of runway must be sufficient to accommodate the aircraft at its maximum takeoff and landing weights.

ii. Airport elevation:

- As the elevation increases, the air becomes less dense. This requires longer runways to achieve the required speed for takeoff.
- For every rise of 300 meters elevation, a runway's basic length is increased by 7%.

iii. Air temperature:

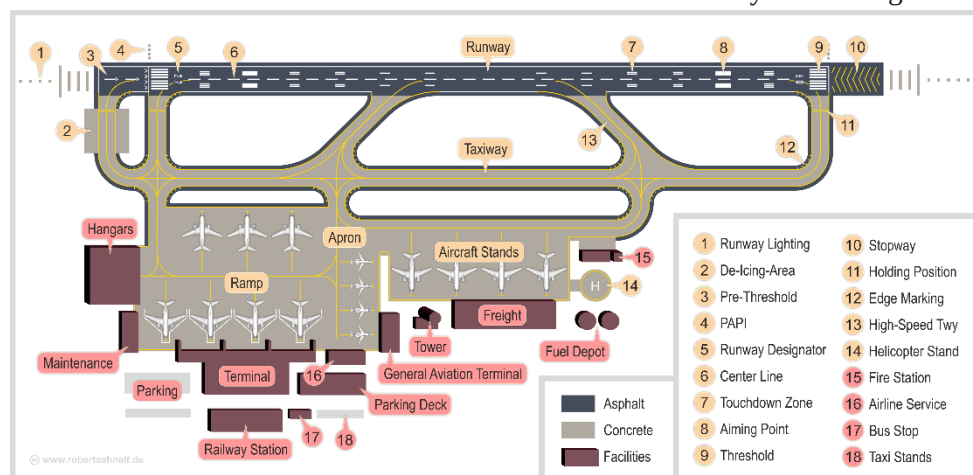
- As the temperature increases, air density becomes low. This develops the need of longer runways reducing life and thrust.
- For every rise of 1°C for which the aerodrome reference temperature exceeds the standard atmospheric temperature, the length of runway increases by 1%.

iv. **Gradient of runway:**

- Longer runway is required for takeoff in an uphill gradient of runway whereas a downhill gradient requires lesser length.
- The effect of runway gradient varies with the weight of the aircraft and ambient air temperature.
- For turbine-powered aircraft, runway length is affected by **7 to 10% for each 1% of uniform gradient**.
- For planning purposes, the FAA uses an effective gradient which is the difference in elevation between the highest and lowest points on the runway profile divided by the length of the runway.

v. **Wind conditions:**

- Shorter length of runway is required if headwind is greater.
- Longer runway is required for greater tailwind.
- Crosswinds also increase the runway length required.



Holding bays:

A defined area near runway ends where aircraft park briefly before takeoff is known as holding bays. The need of Holding Bays in airport is listed below:

i. **Management of Runway:**

Aircraft wait for the clearance of runway when runway is busy. This prevents the possibility of aircraft accidents due to collision.

ii. **Check routines prior to takeoff:**

Piston-engine aircraft perform engine warm-up and instrument checks before takeoff in holding bays without holding the airway traffic to carry out other takeoff and landing operations.

iii. **Takeoff based on priority:**

Aircraft can take off in ATC-cleared priority order rather than queuing on taxiways.

iv. **Flexible departure:**

Aircraft can leave and independently re-enter the departure area.

v. **Emergency handling:**

If one aircraft cannot take off, others can bypass it.



Bypasses:

A defined area on the airfield where aircraft in motion either taxiing or towing can be held to facilitate the ground movements is known as bypass.

- i. **Flexibility in operation:**
Second taxiway/ bypasses provide alternative routes at relatively low construction cost for allowing the ground movement of aircraft without keeping the air traffic at hold.
- ii. **Reduction in delays:**
Aircraft can bypass parked/holding aircraft without waiting which reduces the frequency of delays of aircraft.
- iii. **Prevention of blast interference:**
To prevent engine blast to affect the passing aircraft, aircraft are placed in bypass routes for this operation.
- iv. **High traffic volume:**
Bypasses are provided when traffic volume is high such that it exceeds capacity during busy periods.

Section-D (25 Marks)

11. Answer the following:

[4+6=10]

- a. Imagining that a mayor of a sub-metropolitan city in Nepal ask you to provide a Lum sump cost required to construct a new water supply system to serve the city's population. Making reasonable assumptions of the per capita cost and other variables including transmission line, distribution line and treatment units, suggest a tentative figure.
- b. If the above system uses river water for the water supply project, suggest a schematic flow diagram illustrating the possible treatment units involved in the water treatment process. Explain the reasoning behind the sequence of these units.

Answer:

Let us consider that the proposed sub-metropolitan city has:

Population to be served = **100,000 persons**

Per capita water demand = **135 LPCD**

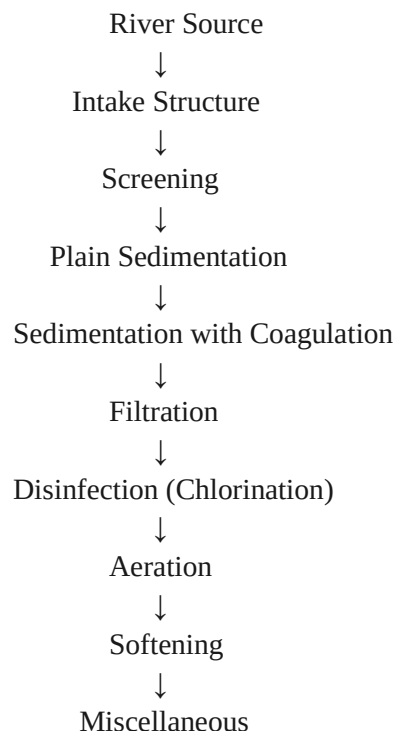
Total demand = $100,000 \times 135 = 13.5 \text{ MLD}$

Design period = 20 years

Component	Assumed Cost (NRs.)
Source and intake work	80 million
Water treatment plant (13.5 MLD)	250 million
Transmission main	200 million
Distribution network	450 million
Reservoirs and service tanks	150 million
Pumps, electromechanical works	70 million
House connections and meters	100 million
Engineering, supervision and contingencies (10%)	130 million
Total Estimated Cost	1.43 billion NRs.

Hence, we can conclude that approximate Lump sum cost required to construct a new water supply system to serve the city's population to be 1.4-1.5 billion. This figure may vary depending on topography, source distance, treatment requirements, and local construction costs.

Schematic flow diagram illustrating the possible treatment units involved in the water treatment process:



Intake structures:

- Hydraulic device constructed at or near river source to draw the required amount of water from the river source.
- Intake structures must be provided with the gate system to prevent entry of floating debris, large suspended sediments to the treatment plant.

Screening:

- Screening is the process of passing water through screens to remove large floating and suspended matter such as sticks, branches of tree, dead animals etc.
- Depending upon the size of the screens, there are three types of screens: coarse screen, fine screen and medium screen.
- Screening reduces the load on the treatment plant and increases the efficiency of treatment.

Plain sedimentation:

- The process of removal of coarse suspended particles present in the water such as sand, silt by allowing it to remain quiescent in sedimentation tank so that the suspended material settle down at the bottom by the action of gravity is known as plain sedimentation.
- Detention period in a plain sedimentation tank= 4-8 hours.
- SOR of a plain sedimentation tank=15 to 30 m³/m²/day

Sedimentation with coagulation:

- The fine suspended and colloidal materials like clay particles which cannot be removed by plain sedimentation or takes long time to removal by plain sedimentation can be removed by sedimentation with coagulation.
- Coagulants such as aluminum sulphate, sodium aluminate, iron salts are used.
- Detention period for sedimentation with coagulation = 1-4 hours

- SOR for sedimentation with coagulation= 30 to 40 m³/m²/day

Filtration:

- Filtration is the process of passing water through the beds of sand and granular material.
- For the removal of those impurities whose specific gravity is less than or equal to that of water for which sedimentation process does not work, in such case filtration is used.
- Slow sand filter, Rapid sand filter, pressure filters are some of the common filters used for the filtration of water in treatment plants.

Disinfection:

- The process of killing pathogenic bacteria and microorganisms present in water for the safe use is known as disinfection.
- The chemicals or substances used to kill microorganisms are disinfectants.
- Chlorination is one of the most common processes of disinfecting water.

Aeration:

- The process of bringing water in contact with air to remove dissolved gasses present in water is known as aeration.
- Aeration removes CO₂ gasses up to 70% due to which corrosion corrosiveness of water is reduced.

Softening:

- The process of removal or reduction of temporary or permanent hardness of water is known as softening.
- Softening helps to improve the taste of water, reduce consumption of soap during washing, bathing and also reduces formation of scales in boiler.

Miscellaneous:

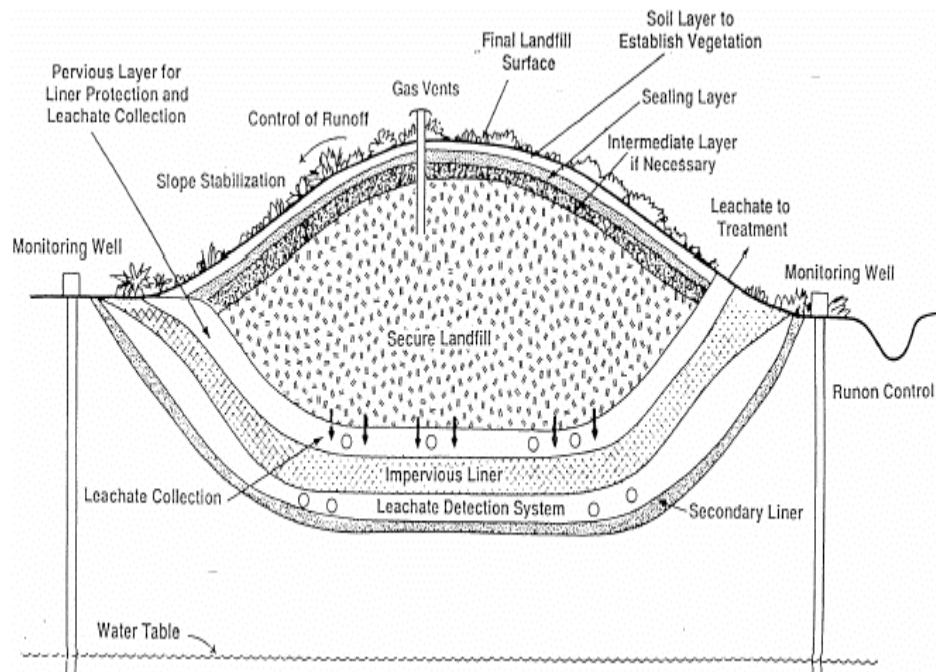
- It includes:
 - Removal of iron and manganese
 - Removal of color, odor, taste from the water
 - Removal of arsenic and so on.

The treatment units are arranged in such a way that it progressively remove large debris, suspended solids, colloidal particles, microorganisms, and pathogens, to ensure that the water meets drinking water quality standards before distribution to consumers.

12. What are the key components of a sanitary landfill site designed to dispose solid waste with minimal environmental and health risks? Elaborate with a neat sketch. [10]

Answer:

A sanitary landfill is an engineered facility for the safe disposal of solid waste with minimum environmental pollution and public health hazards. A sanitary landfill is characterized by the **presence of a liner, leachate collection system** which prevents groundwater contamination and a **capping system** that prevents air contamination. Sanitary landfill is different from ordinary landfill as open dumps lack these engineering controls.



The key components of sanitary landfill are as follows:

i. Bottom liner system:

- Prevents leachate to contaminate groundwater.
- It consists of compacted clay layer that act as a barrier reducing permeability.
- Bottom liner system may also be provided with geomembrane (HDPE pipes) for preventing leachate seepage into the ground.

ii. Leachate collection system:

- This system collects leachate generated from waste decomposition and rainwater infiltration.
- Prevents groundwater contamination due to wastages.
- It consists of:
 - Drainage layers: Allows the flow of leachate to perforated pipes
 - Perforated pipes: provided at the base of landfill to collect the contaminated liquid.
 - Collection sumps: storage of leachate before treatment.
 - Treatment facilities: treatment of leachate.

iii. Landfill gas collection system:

- This system collects gases mainly methane and carbon dioxide produced during decomposition.
- It prevents explosions and air contamination.
- The collected gas can also be used for energy generation.
- Consists of vertical and horizontal gas wells to extract methane from decomposing waste, Pipes and manifolds to transport gas to processing facilities.

iv. Capping system:

- It is the cover system applied after the landfill reaches its design capacity.
- It consists of clay, geomembrane, drainage layer and topsoil with vegetation.
- Capping system helps in minimizing rainwater infiltration and controlling gas emissions.

In Nepal, one of the most talked about landfill site is **Banchare Danda landfill site**, serving as the permanent disposal site for the Kathmandu Valley receiving about 1500 metric tons of daily wastages. Locals of the area frequently block garbage trucks due to unfulfilled government agreements, foul odors, and severe environmental/health hazards. Another landfill site located nearby Kathmandu is **Sisdol landfill site** which was intended to be a 3-year temporary dump, but operated for almost two decades. Enormous quantity of wastage

accumulation on the area resulted in groundwater contamination and disease. The situation led to years of protests before the shift to Banchare Danda.

From these examples, we can conclude that sanitary landfill is one of the most eco-friendly method of disposal of waste that not only prevents the groundwater contamination by leakage of leachates into the water but also encourages sanitized and hygienic practice of waste disposal. Practice of waste separation from the generators also help in proper waste management. Also, the quantity of wastages generated from each household and enterprises must be reduced by adopting the concept of recycle and reuse. Different organizations such as Doko Recyclers, smartpaani, SAHAS Nepal are also working actively in grassroot level to encourage people for adopting alternative water recycling practices such as greywater management, eco-friendly waste collection etc